Filippo Ricci

Mathematics — Partial Differential Equations & Calculus of Variations

filippo.ricci@gssi.it (+39) 366 139 7435 filippooricci.github.io Rome, Italy

Research Profile

Master's graduate in Mathematics from the University of Trieste, with a curriculum in Advanced Mathematics and a strong background in partial differential equations, calculus of variations, functional analysis, geometric measure theory, and hyperbolic conservation laws.
My Master's research focuses on the construction of a global weak solution to the shallow-water system in a rotating frame through a variational time-discretization scheme based on the minimization of the mechanical energy of the system.
I have been awarded a PhD scholarship in Mathematics at the Gran Sasso Science Institute (GSSI), where I began my doctoral studies in November 2026 under the supervision of Prof. Sara Daneri. My position is funded within the FIS3 project SEASTAR (*SElf-ASsembled paTterns: Analysis and Regularity*).

Education

M.Sc. in Mathematics Sep. 2024 – Jul. 2026
University of Trieste *Trieste, Italy*

- Final grade: **110/110 cum laude**.
- Curriculum: *Advanced Mathematics*.
- Master's thesis: *A Variational Approach to the Shallow-Water Equations via an Energy Minimization Process*.
- Supervisor: **Prof. Stefano Bianchini**, SISSA — International School for Advanced Studies.

Diploma, Percorso Formativo Comune Sep. 2024 – Jul. 2026
International School for Advanced Studies *Trieste, Italy*

B.Sc. in Mathematics Aug. 2021 – Oct. 2024
Sapienza University of Rome *Rome, Italy*

- Final grade: **110/110**.
- Curriculum: *Generale*.
- Bachelor's thesis: *Insiemi di Perimetro Finito* (*Sets of Finite Perimeter*).
- Supervisor: **Prof. Annalisa Malusa**, Sapienza University of Rome.

Awards & Scholarships

PhD Scholarship in Mathematics 2026
Gran Sasso Science Institute (GSSI), L'Aquila, Italy
Awarded through competitive public selection. The scholarship is funded within the FIS3 project **SEASTAR** – **SElf-ASsembled paTterns: Analysis and Regularity**, led by Prof. Sara Daneri.

SISSA Scholarship — Percorso Formativo Comune 2024–2026
SISSA — International School for Advanced Studies / University of Trieste
Awarded a competitive scholarship to attend advanced Master's-level courses within the *Percorso Formativo Comune*, jointly organized with SISSA and the University of Trieste.

Research Experience

Master's Thesis Research University of Trieste / SISSA
A Variational Approach to the Shallow-Water Equations via an Energy Minimization Process

- Studied the shallow-water system in a rotating frame, with the aim of constructing global weak solutions.
- Developed a variational time-discretization scheme in which, at each time step, the evolution is determined through the minimization of the mechanical energy of the system.
- The variational construction is inspired by the optimal transport approach introduced by Gangbo and Westdickenberg for the system of isentropic Euler equations.

- Used tools from partial differential equations, calculus of variations, and optimal transport to construct a global weak solution to the system.

Supervisor: Prof. Stefano Bianchini (SISSA, Trieste).

Selected Graduate Coursework

Functional Analysis
Calculus of Variations
Hyperbolic Conservation Laws
Partial Differential Equations
Geometric Measure Theory
Stochastic Calculus

Teaching Experience

University Tutor Oct. 2025 – Dec. 2025
University of Trieste *Trieste, Italy*

- Provided tutoring sessions for first-year university students.
- Prepared weekly problem sets and exercises.

Private Tutor in Mathematics and Physics Nov. 2021 – May 2022
Private teaching *Rome, Italy*

Provided after-school mathematics and physics tutoring to secondary-school students.

Technical Skills

Programming C++, MATLAB
Office software Microsoft Word, Excel, PowerPoint

Languages

Italian Native
English B2 — listening, reading, writing, spoken production and interaction

Upcoming Position

PhD in Mathematics Nov. 2026 –
Gran Sasso Science Institute (GSSI) *L'Aquila, Italy*

Awarded a PhD scholarship in Mathematics through the competitive public selection procedure. Doctoral studies will begin on **2 November 2026**.

Additional Activities

Youth Education and Volunteering Rome, Italy

Long-term involvement in educational and recreational activities for children and teenagers, including catechesis, summer camps, activity planning, and logistical organization.

Sports

Football training and participation in official FIGC youth competitions; regular gym and swimming activities.